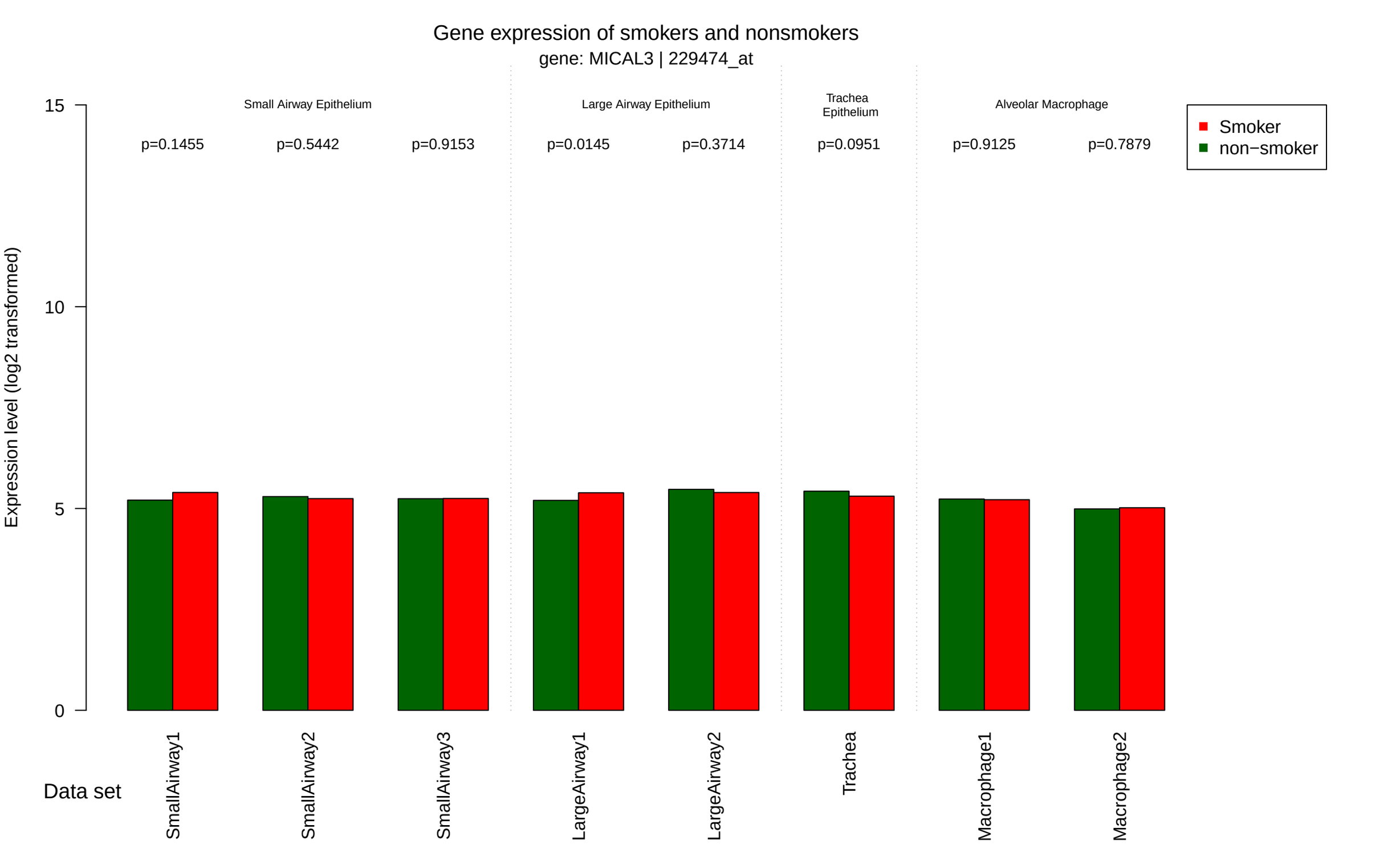
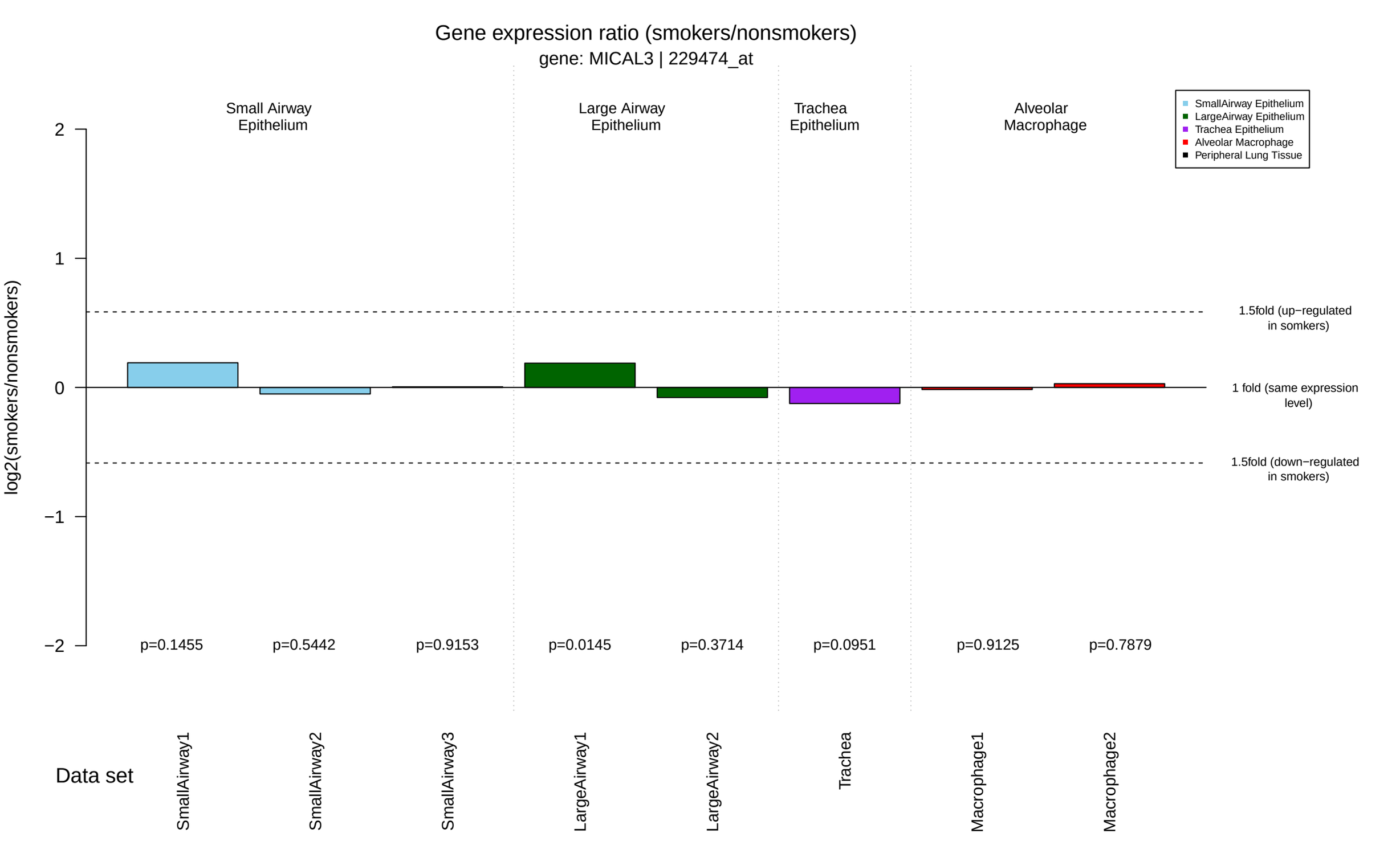
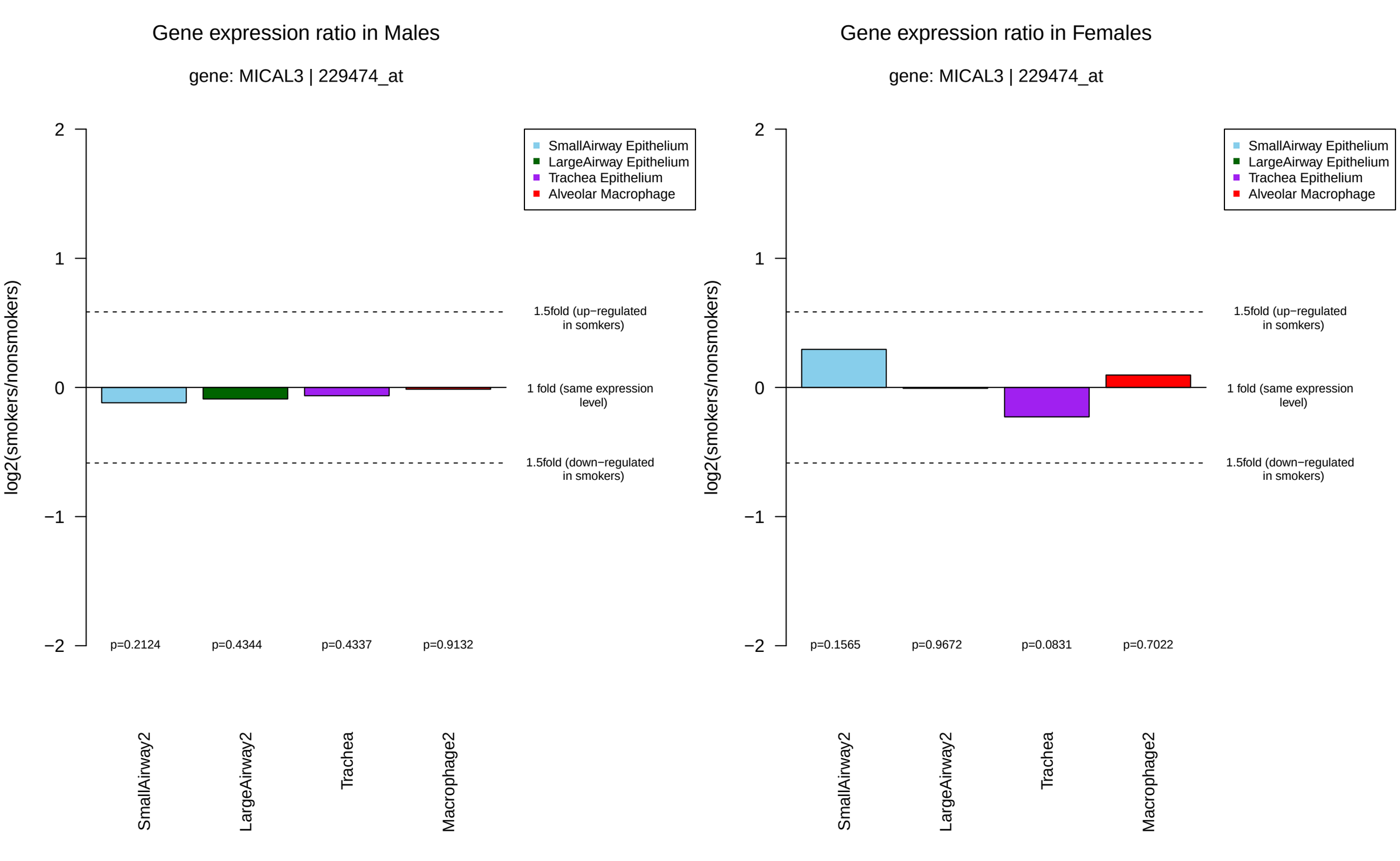
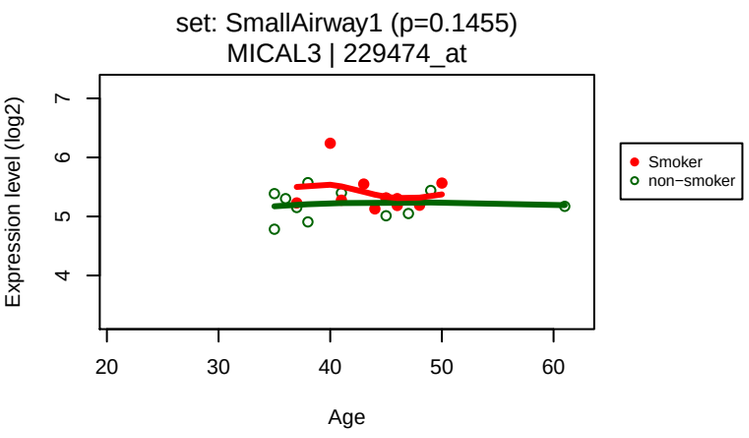
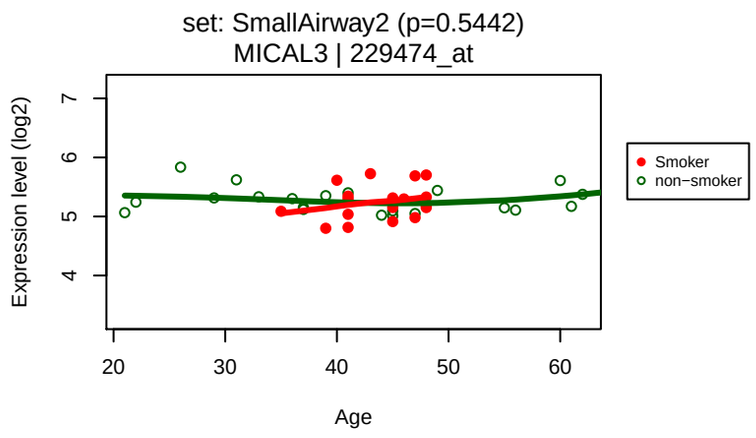
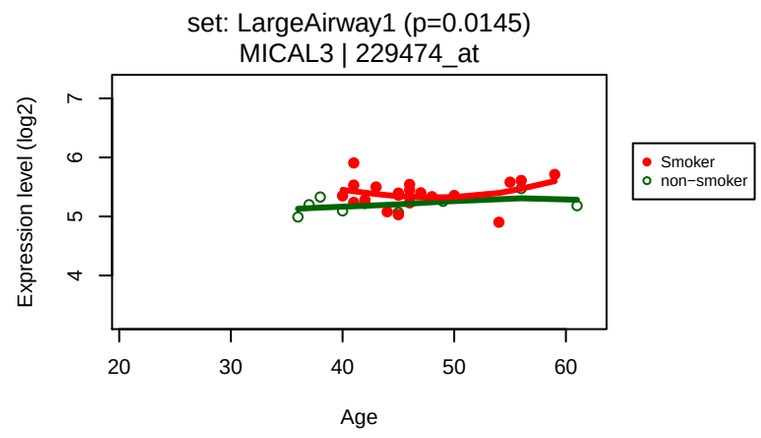

Gene expression of smokers and nonsmokers
gene: MICAL3 | 229474_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1455
p=0.5442
p=0.9153
p=0.0145
p=0.3714
p=0.0951
p=0.9125
p=0.7879
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

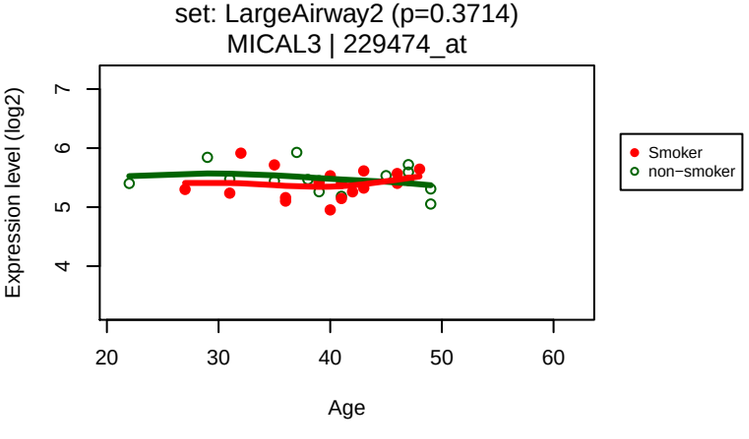

Gene expression ratio (smokers/nonsmokers)
gene: MICAL3 | 229474_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1455
p=0.5442
p=0.9153
p=0.0145
p=0.3714
p=0.0951
p=0.9125
p=0.7879
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

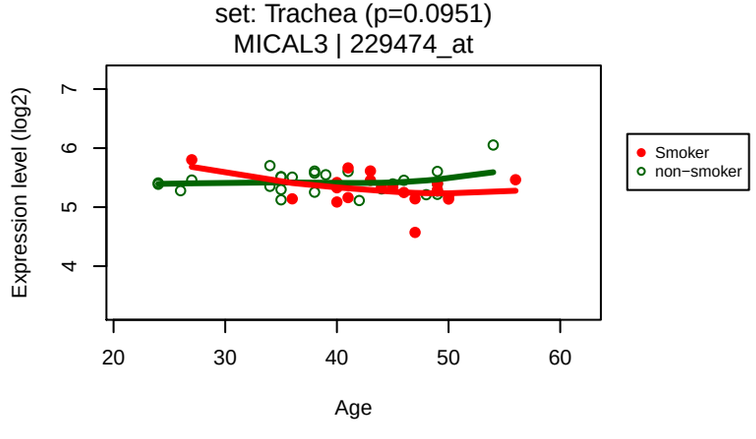

Gene expression ratio in Males
gene: MICAL3 | 229474_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2124
p=0.4344
p=0.4337
p=0.9132
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MICAL3 | 229474_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1565
p=0.9672
p=0.0831
p=0.7022
SmallAirway2
LargeAirway2
Trachea
Macrophage2

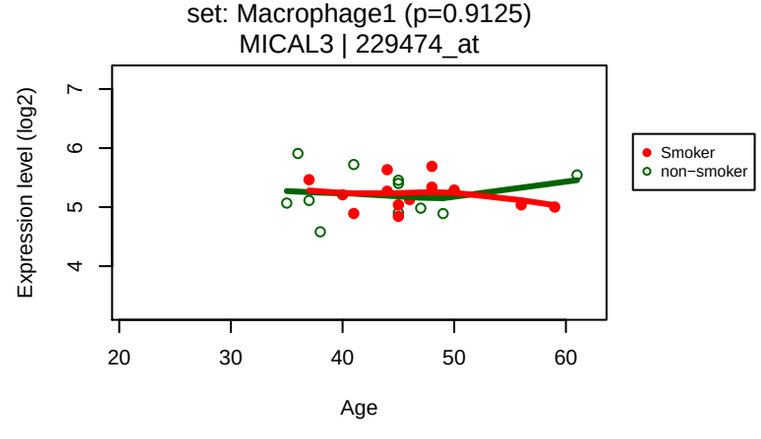

set: SmallAirway1 (p=0.1455)
MICAL3 | 229474_at
Expression level (log2)
Age
Smoker
non−smoker

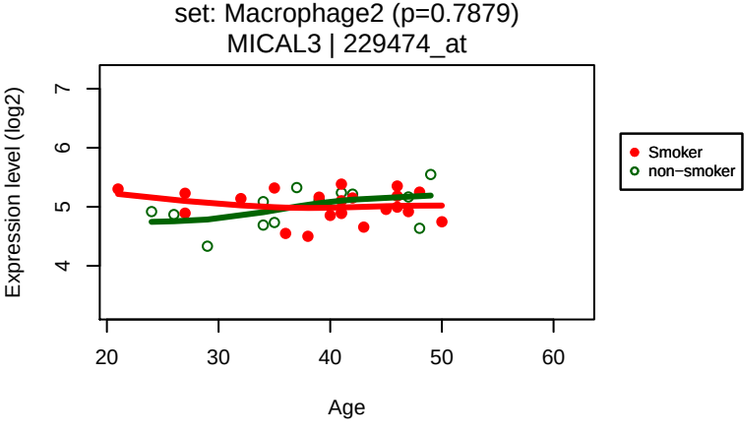

set: SmallAirway2 (p=0.5442)
MICAL3 | 229474_at
Expression level (log2)
Age
Smoker
non−smoker

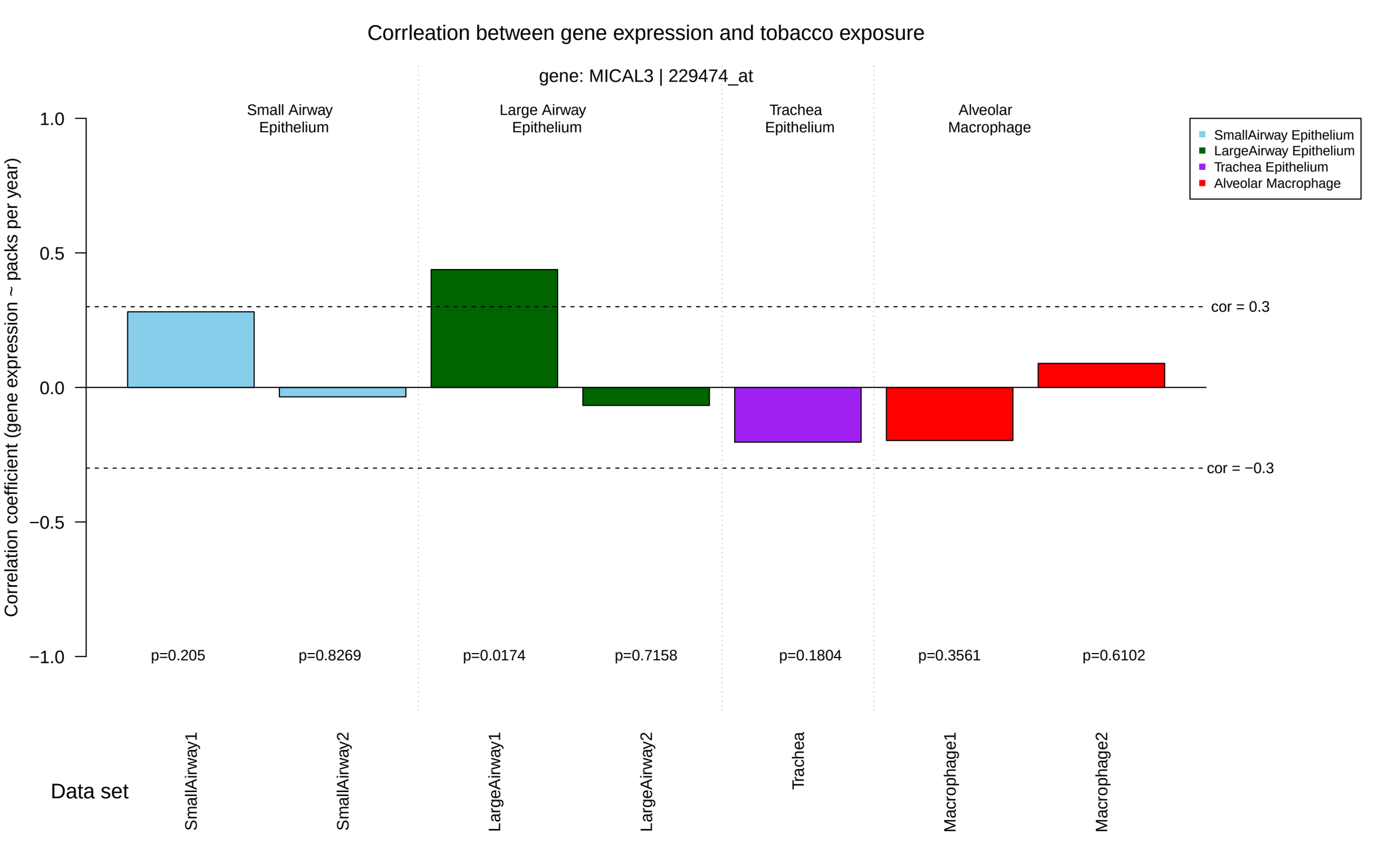

set: LargeAirway1 (p=0.0145)
MICAL3 | 229474_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3714)
MICAL3 | 229474_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0951)
MICAL3 | 229474_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9125)
MICAL3 | 229474_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7879)
MICAL3 | 229474_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MICAL3 | 229474_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.205
p=0.8269
p=0.0174
p=0.7158
p=0.1804
p=0.3561
p=0.6102
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2